



Hall Ticket No. -

Code: S123536

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME) III Semester (Backlog) Examinations-May-June-2023

Digital Electronics

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

- a. State De Morgan's theorem. [2M CO1 L1]
- b. Construct 4X1 multiplexer. [2M CO2 L1]
- c. Describe an example of 2-input lookup table. [2M CO3 L1]
- d. Recite transition table of JK Flip-Flop. [2M CO4 L1]
- e. Recall basic design steps of a Mealy machine. [2M CO5 L1]
- f. Draw F.A using H.A circuit. [2M CO2 L3]

Part-B

4X12=48M

- a) Minimize the following Boolean expression using K-map
 $f = m(0,1,2,3,5,6,7,8,9,10,13,14,15)$. [6M CO1 L3]
- b) Implement the function $F = X'Y + XY'$ using
(a) only NAND gates and (b) only NOR gates. [6M CO1 L3]
3. a) Design binary to gray code converter and implement with NOR gates. [6M CO2 L2]
- b) Design and explain 3 x 8 Decoder with neat logical circuit diagram. [6M CO2 L2]
4. a) Break down implementation of Boolean functions $F1 = XY'Z' + X'Y' + Y'Z$ and
 $F2 = XYZ + Y'Z$ by using Programmable Logic Array device. [6M CO3 L4]
- b) Outline the functioning of optimized 3 bit odd parity generator using PAL. [6M CO3 L4]
5. a) Explain the significance of excitation table with an example. Differentiate between
combinational and sequential circuits. [6M CO4 L5]
- b) Design MOD 6 synchronous counter using T flip-flops. [6M CO4 L5]
6. a) Demonstrate Moore machine with example. Design a mealy machine to find 2's
complement of a number. [6M CO5 L6]
- b) Demonstrate state minimization with an example. Design a non-overlapping sequence
detector to detect an input sequence of '101' [6M CO5 L6]
7. a) Design SISO, PISO, Universal shift register. [6M CO4 L2]
- b) Explain full adder and full subtractor with a neat diagram. [6M CO1 L2]

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Part-A

6X2=12M

- 1 a Define probability mass function and density function. 2M CO1 BT1
- b State uniform distribution. Hence write mean and variance of uniform distribution. 2M CO2 BT2
- c Write normal equations for fitting of a straight line by the method of least squares. 2M CO3 BT2
- d Determine the value of $y(0.2)$ by Euler's method for $\frac{dy}{dx} = x + y$, $y(0) = 1$ with step size $h = 0.2$. 2M CO4 BT3
- e Find whether the set of vectors $(1, 2, 3)$, $(4, 5, 6)$ and $(5, 7, 9)$ are LI or LD 2M CO5 BT3
- f Write any two properties of normal distribution. 2M CO2 BT2

Part-B

4X12=48M

- 2 a There are three bags: first containing 1 white, 2 red, 3 green balls ; second 2 white , 3 red, 1 green balls and third 3 white , 1 red , 2 green balls. Two balls are drawn from a bag chosen at random. These are found to be one white and one red. Find the probability that the balls so drawn came from second bag. 6M CO1 BT3

- b A random variable X has the following probability function :

6M CO1 BT3

X	0	1	2	3	4	5	6
P(x)	K	3k	5k	7k	9k	11k	13k

Find i) Value of k ii) $P(x < 3)$ iii) $P(2 < x < 5)$.

- 3 a In a test on 2000 electric bulbs, it was found that the life of a particular make was normally distributed with an average life of 2040 hours and standard deviation of 60 hours. Estimate the number of bulbs likely to burn for i) more than 2150 hours ii) less than 1950 hours iii) more than 1920 hours and less than 2160 hours. 6M CO2 BT3

- b X is continuous random variable with probability density function given 6M CO2 BT3

$$\text{by } f(x) = \begin{cases} kx & 0 \leq x < 2 \\ 2k & 2 \leq x < 4 \\ -kx + 6k & 4 \leq x \leq 6 \end{cases} \text{ the find } k, \text{ mean and variance}$$

- 4 a Test the equality of means at 5% LOS for the given samples.

6M CO3 BT3

S1	17	26	35	40	8	9	11
S2	9	14	2	3	6		

- b Fit a second degree parabola to the following data by the method of least squares:

6M CO3 BT3

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	1.1	1.3	1.6	2.0	2.7	3.4	4.1

- 5 a Apply Runge-Kutta method to find approximate value of y when $x = 0.2$ given that $\frac{dy}{dx} = x + y$, $y(0) = 1$ with step size $h = 0.1$.

6M CO4 BT3

- b Given that $\frac{dy}{dx} = x^3 + y$ where $y(0) = 2$, $y(0.2) = 2.073$, $y(0.4) = 2.452$ and $y(0.6) = 3.023$ then find $y(0.8)$ by Milne's predictor corrector method with step size $h=0.2$.

6M CO4 BT3

- 6 a Find the singular value decomposition of $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$

12M CO5 BT3

- 7 a The two regression equations of the variables x and y are $\bar{x} = 19.13 - 0.87y$ and $y = 11.64 - 0.50\bar{x}$.

6M CO3 BT3

- i) Find mean of x's
ii) Find mean of y's
iii) the correlation coefficient between x and y.

- b Examine whether the vector $v = (3, 7, -4) \in R^3$ is linear combination of vectors $u_1 = (1, 2, 3)$, $u_2 = (2, 3, 7)$ and $u_3 = (3, 5, 6)$ which are in R^3 or not.

6M CO5 BT4

Stanley College of Engineering & Technology for Women (A)**B.E (CSE) III Semester (Backlog) Examinations-May-June-2023****Computer Organization****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
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iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. Identify the importance of program counter. [2M CO2 BTL1]
- b. List I/O Reference Instructions [2M CO1 BTL2]
- c. List the advantage of micro-programmed control unit. [2M CO3 BTL1]
- d. What is Hit ratio? [2M CO2 BTL4]
- e. Categorize the factors considered in designing an I/O subsystem. [2M CO5 BTL4]
- f. Describe the addressing modes supported by 8051? [2M CO4 BTL1]

Part-B**4X12=48M**

2. a) Explain the different functional units of a computer. [6M CO2 BTL2]
- b) Write the zero and three address instructions using RTL for the following arithmetic statement $X = (A + B) * (C + D)$. [6M CO1 BTL5]
3. a) Illustrate the instruction cycle with neat diagram. [7M CO1 BTL3]
- b) Explain about memory reference instructions. [5M CO1 BTL2]
4. a) Explain Virtual Memory in detail. [4M CO2 BTL2]
- b) Identify the various possible hazards that occurs in the instruction pipeline and find the solutions to solve it. [8M CO3 BTL4]
5. a) Illustrate the pin diagram of 8085 microprocessor and explain it. [6M CO4 BTL2]
- b) Illustrate with neat diagram, how the daisy chaining priority solve the problem of interrupt. [6M CO5 BTL3]
6. a) Explain cache memory in detail. [6M CO4 BTL3]
- b) Describe the computer registers in detail. [6M CO4 BTL2]
7. a) Draw the block diagram of DMA controller and explain it. [6M CO5 BTL2]
- b) List out the different addressing modes and explain it. [6M CO2 BTL2]



Hall Ticket No. -

Code: S1235315

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME/AI&DS/IT) III Semester (Backlog) Examinations-May-June-2023

OOPS Using Java

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

- a. Define Class and Object. [2M CO1 BTL1]
b. What is abstraction in java? [2M CO2 BTL1]
c. What are the different ways of thread creation? [2M CO3 BTL1]
d. What are exception handling keywords? [2M CO3 BTL1]
e. Define Serialization in Java. [2M CO4 BTL1]
f. Define event source and event listener. [2M CO5 BTL1]

Part-B

4X12=48M

- a) What is Scope and life time of a variable? Explain with an example. [6M CO1 BTL4]
b) Write a program to demonstrate Class, Object and Method. [6M CO1 BTL3]
- a) Difference abstract classes and interfaces. [4M CO2 BTL2]
b) Write a program to demonstrate multilevel inheritance. [8M CO2 BTL4]
- a) Explain exception handling in Java with an example. [6M CO3 BTL2]
b) Write a program to demonstrate thread creation using runnable interface. [6M CO3 BTL3]
- a) Explain the collection class hierarchy. [7M CO4 BTL3]
b) Demonstrate Scanner class with an example program. [5M CO4 BTL3]
- a) Explain the Delegation Event Model in AWT. [6M CO5 BTL2]
b) What are the different types of event listeners supported by Java? Explain. [6M CO5 BTL2]
- a) Demonstrate constructor overloading concept with the help of a program. [7M CO1 BTL2]
b) What is the use of adapter classes? Explain. [5M CO5 BTL2]
